



# Transaction fraud detection via attentional spatial–temporal GNN

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## Abstract

The detection of fraudulent transactions remains a critical challenge in the financial services industry, further intensified by the rapid growth in transaction volumes and the increasingly sophisticated tactics of fraudsters. Traditional methods, including current graph neural networks, have shown potential but often fall short in accurately detecting fraud due to the ability of fraudsters to mimic legitimate behaviors, conceal relationships, and operate within narrow time frames. These limitations necessitate the development of more effective detection techniques. This study is motivated by the need to overcome the inherent limitations of existing GNN-based methods in fraud detection. We propose an innovative approach that enhances local information aggregation and incorporates global insights through graph reconstruction, addressing both the structural and temporal complexities introduced by fraudulent activities. Our method utilizes attention mechanisms to capture spatial–temporal relationships within transaction networks, emphasizing crucial connections between nodes to improve detection efficiency. Key innovations include attentional intra-relation aggregation, which differentiates the strength of connections between neighboring nodes, and attentional inter-relation aggregation, which prioritizes nodes that share a higher number of common neighbors. We validate the effectiveness of the proposed method using two benchmark datasets, YelpChi and Amazon, as well as a real-world dataset. Experimental results demonstrate that our approach significantly outperforms existing techniques, achieving notable improvements in AUC, F1-macro, and Gmean metrics. These results highlight the importance of addressing the current limitations in GNN-based methods and demonstrate the potential of our approach to provide a more robust solution for financial fraud detection.

**Keywords** Graph neural network · Fraud detection · Spatial–temporal attentional graph neural network · Attentional intra-relation aggregation · Attentional inter-relation aggregation · Camouflage detection

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## 1 Introduction

Fraudulent activities, particularly in credit card transactions, pose significant challenges in financial services, leading to substantial financial losses [1, 2]. The urgency of identifying fraudulent transactions is paramount, as it helps reduce financial losses, enhance customer experience, and safeguard the reputation of financial institutions. Various techniques have been proposed to detect fraud, broadly categorized into rule-based methods and machine learning-based methods.

Rule-based methods rely on expert-generated rules to evaluate fraud likelihood but often fall short in complex environments due to their rigidity and dependence on specialist knowledge [3]. In contrast, machine learning-based methods leverage historical transaction data to identify fraudulent activities, typically extracting features such as statistical indicators from transaction characteristics like time, amount, and location. However, these methods struggle to capture intricate interactions between transactions and often overlook the spatial and temporal relationships among them, presenting significant challenges for effective fraud detection [3].

Recent advancements highlight the role of graph-based techniques, particularly graph neural networks (GNNs), in enhancing fraud detection capabilities. GNNs aggregate information from neighboring nodes to generate representations for central nodes, facilitating automatic feature learning that considers interactive connections between transactions [4–6]. Despite their potential, GNN-based methods encounter challenges in learning spatial–temporal information effectively. Fraudsters often operate using limited devices due to high equipment costs, making spatial aggregation crucial. Additionally, fraudulent actions frequently occur within tight time frames, as alerting cardholders can lead to transaction stoppage [3].

Current GNN-based detection methods primarily focus on local information from neighboring nodes for updating target node representations while neglecting the global information essential for comprehensive fraud detection. This limitation can lead to over-smoothing, where the learned representations become homogeneous, reducing the detection capabilities of GNN-based fraud detectors. Therefore, there is a pressing need for methods that leverage both global information and long-path transactions to mitigate over-smoothing and improve detection accuracy [7].

Another significant challenge is node camouflage, where fraudsters mimic benign behavior to evade detection. This tactic, exacerbated by the prevalence of benign nodes, complicates the identification of fraudulent nodes and hinders the effective use of individual fraud features. Current camouflage detection approaches struggle to account for the graph's spatial structure while filtering nodes exhibiting camouflage behavior [8–10]. Existing methods, such as the one proposed in [7], attempt to address these challenges by reconstructing subgraph structures using higher-order adjacency matrices; however, they often overlook connection intensity within the reconstructed adjacency matrix [11–13].

To address these limitations, the proposed method combines GNNs with attentional mechanisms to identify fraudulent transactions. Initially, it examines

hidden connections among detected fraudsters via long-distance links in the node's adjacency matrix. The graph auto-encoder (GAE) reconstructs the adjacency matrix of fraudulent nodes, revealing potential connections among them. A temporal encoding strategy is integrated into the GNN architecture for effective temporal information aggregation, followed by a spatial aggregation layer that collates neighboring information while considering temporal differences among transactions.

Our approach introduces attentional intra-relation aggregation, focusing on potential connections among neighboring nodes, and attentional inter-relation aggregation, prioritizing nodes with a higher number of shared neighbors with the target node. A transformer sub-network is incorporated at the top of the GNN layer stack to enhance the model's capacity for understanding global and structural information, followed by a classifier to categorize nodes. Extensive experiments on benchmark datasets (Amazon and YelpChi) and a real-world dataset validate the effectiveness of the proposed method, which outperforms existing fraud detection techniques.

The primary innovations of this study include: I) a novel method for identifying fraudulent transactions that learns spatial–temporal information while preserving structural information; II) attentional intra-relation aggregation, leveraging attention mechanisms for potential connections among neighboring nodes; III) attentional inter-relation aggregation, integrating global information into local aggregation; and IV) experimental results that demonstrate the proposed method's effectiveness across multiple datasets.

The remainder of the paper is organized as follows: Sect. 2 reviews the relevant research background. Section 3 details the materials and methods employed in this study. Section 4 presents the datasets, evaluation metrics, and experimental results. Finally, the conclusion is provided in Sect. 5.

## 2 Research background

The related work is summarized in three main areas: (1) GNN-based methods for fraud detection; (2) transformers in GNNs for fraud detection; and (3) spatial–temporal GNNs.

### 2.1 GNN-based methods for fraud detection

Fraud detection can be efficiently accomplished through node analysis leveraging graph neural networks. This method uncovers fraudulent nodes by examining their associations and interactions with adjacent nodes. Frequently, fraudsters take advantage of ties with legitimate users and offer what appears to be credible feedback to support their deceitful endeavors [7, 12].

A novel fraud detection method improves the efficacy of GNNs by tackling two significant obstacles: sparse connections and multi-relational data [14]. SCN\_GNN enhances fraud detection by merging robust node information with graph structure.

The approach incorporates the structured similarity-aware module (SSAM) for elevating resolution and the strong node module (SNM) for diminishing it.

In [7], a two-phase GNN-based method, referred to as CIES-GNN, was developed to tackle camouflage challenges in fraud detection and improve semantic aggregation in diverse networks. This method reconstructs subgraphs to reveal concealed fraudulent links and removes unnecessary connections between fraudulent and legitimate nodes. Furthermore, it employs an enhanced semantic aggregation module to amalgamate higher-order information across various relationships, thus improving node embeddings while preventing over-smoothing.

In [15], the latent synergy graph learning-based telecom fraud detector (LSG-FD) is given, which can detect both sequential and interactive fraudulent actions inside graph-structured data. This model addresses network sparsity, a lack of synergistic relationships among fraudsters, and graph disassortativity. To solve these issues, LSG-FD reconstructs latent synergy graphs and uses an LSTM-based behavior encoder to evaluate calling patterns.

In [12], the HA-GNN model is introduced. This groundbreaking method addresses the challenge of fraudsters using deceptive tactics, such as creating fake connections with authentic users and providing feedback that seems genuine. HA-GNN operates through three main stages: initially, it encodes each node with the help of adjacency matrices; next, it employs multi-head self-attention to capture comprehensive structural information; and finally, it produces node embeddings by integrating structural data with feature details.

In [16], the Batch and Real-time Inception Graph Topology (BRIGHT) system is presented as an innovative method for implementing GNNs in real-time fraud detection. It tackles two primary challenges: eliminating reliance on future data for predictions and minimizing inference latency. The BRIGHT framework employs a two-stage directed graph to guarantee that only historical transactions are utilized for forecasting.

In [17], the GraphFC model presents an innovative strategy for detecting customs fraud by employing GNNs to tackle the challenge of label scarcity. It converts tabular customs information into a graph format to illustrate transaction connections and applies both self-supervised and semi-supervised learning techniques.

The study [18] emphasizes the considerable gains in fraud detection produced by employing Graph SAmple and aggreGatE (GraphSAGE) and Fast Inductive Graph Representation Learning (FI-GRL). In [19], the Residual Layered Camouflage-Resistant (RLC-GNN) architecture is presented as an enhanced multi-layer GNN for fraud detection. It addresses the constraints of single-layer models such as the CARE-GNN. RLC-GNN improves performance by allowing for progressive learning and error correction through its layered structure and residual connections.

In [10], the pick and choose graph neural network (PC-GNN) is proposed as a GNN-centered approach for tackling the challenges of unbalanced learning in fraud detection. This technique adopts a two-phase strategy to manage imbalanced graph

data: initially, nodes and edges are selected via a label-balanced sampler, followed by the selection of neighbors using a neighborhood sampler.

In [20], xFraud is introduced as an explainable fraud detection technology tailored for extensive transaction networks. It leverages a heterogeneous GNN to forecast fraudulent activities and integrates a hybrid explanation framework to deliver comprehensible insights into the model's predictions.

## 2.2 Transformers in GNNs for fraud detection

Transformer models, initially designed for natural language processing, have been utilized in fraud detection to identify long-range dependencies and contextual insights [21, 22]. The Spatial–Temporal Anomaly Transformer (STAT), discussed in [23], serves as a method for recognizing anomalies in multivariate time series information. STAT utilizes four transformer modules to proficiently capture both spatial and temporal relationships.

Finsformer, as detailed in [24], is an innovative model designed for identifying financial threats, utilizing transformer architecture alongside a distinctive cluster-attention approach. In [25], a technique is introduced that leverages GNN to uncover instances of financial fraud through networks of related party transactions (RPT). While traditional methods often falter due to the intricacies of RPTs, the proposed approach establishes a thorough network by linking corporations with their associated parties and utilizing heterogeneous graph transformers (HGTs) for enhanced detection.

In [26], a decoupled contrastive graph neural network (DC-GNN) is introduced to enhance fraud detection, particularly in cases of camouflage. DC-GNN distinguishes between attribute learning and structure learning to reduce the adverse effects of fraudulent nodes that obscure features or connections.

In [27], the Spatial–Temporal-Aware Graph Transformer (STA-GT) is introduced for detecting transaction fraud. By embedding temporal encoding into GNNs, STA-GT successfully captures the dynamics of spatial and temporal interactions. It also includes a transformer module that acquires insights from both local and global transaction data when addressing long-range interactions.

In [28], a transformer-based approach known as BERT4ETH is introduced for identifying fraud on the Ethereum blockchain. BERT4ETH tackles significant challenges, including transaction duplication, uneven address distribution, and variability in interactions. It employs three primary techniques: reduction of repetitiveness, alleviation of skew, and modeling of heterogeneity, leading to a substantial enhancement in the detection of phishing accounts and the de-anonymization of user accounts.

In [29], a novel distributed knowledge distillation method rooted in the Transformer model is introduced to enhance the detection of financial fraud. This system

improves accuracy by initially assigning feature weights through a multi-attention technique, followed by feature extraction via a feed-forward neural network.

### 2.3 Spatial-temporal GNNs

Incorporating a time component, spatial-temporal graph neural networks provide enhancements to classic graph neural networks. In recent years, various spatial-temporal GNN algorithms have emerged, showcasing exceptional performance in time-sensitive applications compared to other deep learning techniques.

In [3], STAGN, a spatial-temporal attention-based graph neural network, is presented for the detection of credit card fraud. This model utilizes a 3D convolution network to effectively capture and merge spatial and temporal behavioral patterns.

According to [30], spatial-temporal graph boosting networks (STGBNs) are introduced to elevate the predictive efficacy of spatial-temporal graph neural networks (STGNNs) through gradient boosting applications. The STGBN model employs incremental learning, progressively assimilating spatial-temporal data to refine predictions at every step.

In [31], a spatiotemporal constraint graph neural network (StGNN) is developed to generate heterogeneous graphs from medical insurance datasets. This methodology effectively captures intricate spatial and temporal characteristics associated with fraudulent behaviors. StGNN surpasses alternative models by integrating multi-semantic meta-paths and implementing a two-level attention mechanism, thereby enhancing both node classification and anomaly detection.

In summary, in Tables 1, 2, and 3, according to the background of the research, three aspects, including GNN-based methods, transformers in GNNs, and spatial-temporal GNNs, are considered, and the strengths and weaknesses of the models of each aspect are presented.

As observed, GNN-based methods for fraud detection necessitate considerable computational resources owing to the high dimensionality of data and graphs. These approaches tend to be less proficient at detecting fraudulent patterns that involve camouflage or intricate shifts in network relationships. They also face challenges with long-path information aggregation and temporal learning, both of which are essential for effective fraud detection.

Moreover, the integration of transformers within GNNs requires extensive computation and memory during the training phase and may not perform well when fraudulent patterns arise from complex spatial-temporal relationships. In addition, spatial-temporal GNNs demand accurate spatial-temporal data and become inefficient when fraudulent patterns exhibit intricate alterations over time and space.

The proposed method in this article is a new approach based on the use of spatial-temporal information in GNN in order to distinguish between the temporal differences of neighboring transactions in a time slice as well as the dependence between transactions between different time slices. In addition, the use of transformers for

**Table 1** Comparison of the strengths and weaknesses of GNN-based models

| Aspect            | Model        | Strength (ST)/weakness (WE)                                                                                                                                                                                                                                                                                                       |
|-------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GNN-based methods | SCN_GNN [14] | (ST): Enhances fraud detection and classification accuracy in sparse, multi-relational datasets<br>(WE): High complexity limits scalability; noise reduces benefits on dense datasets                                                                                                                                             |
|                   | CIES-GNN [7] | (ST): Excels in imbalanced datasets, detects camouflaged fraud, and enhances node embeddings via higher-order semantic aggregation<br>(WE): High complexity with higher-order adjacency matrices; risk of over-smoothing in multi-relation aggregation                                                                            |
|                   | LSG-FD [15]  | (ST): Detects fraud via synergy links, addresses disassortative graphs, and improves accuracy in imbalanced datasets<br>(WE): High complexity from dual-channel filtering; scalability challenges with larger datasets                                                                                                            |
|                   | HA-GNN [12]  | (ST): Detects camouflaged fraud with hierarchical attention and weighted structural and feature data, improving accuracy<br>(WE): Noise from legitimate connections may impact accuracy; balancing attention between genuine and fraudulent data is challenging                                                                   |
|                   | BRIGHT [16]  | (ST): Prevents future data leaks, ensuring accuracy; reduces inference latency for real-time applications<br>(WE): Relies on historical data, limiting detection of new fraud patterns; complex infrastructure hinders deployment and maintenance                                                                                 |
|                   | GraphFC [17] | (ST): Leverages unlabeled data to improve performance in label-scarce environments and generalizes well to new importers and product categories<br>(WE): Converting tabular data to graph structure adds overhead, and reliance on higher-order interactions increases model complexity                                           |
|                   | FI-GRL [18]  | (ST): Improves fraud detection accuracy with graph-based representations and enhances embeddings through reduced graph-level sampling rates<br>(WE): FI-GRL's reliance on network topology limits effectiveness with few node features, and both methods have high computational costs, impacting scalability for larger datasets |
|                   | RLC-GNN [19] | (ST): Enables deeper layers with progressive learning and residual connections, improving recall, AUC, and F1-score across multiple datasets<br>(WE): Higher complexity increases overfitting risk, particularly on smaller datasets, and requires extensive hyperparameter tuning for optimal performance                        |
|                   |              |                                                                                                                                                                                                                                                                                                                                   |

Table 1 (continued)

| Aspect | Model       | Strength (ST)/weakness (WE)                                                                                                                                                                                                                                                                                         |
|--------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | PC-GNN [10] | (ST): Handles class imbalance well, improving fraud detection for minority classes and showing strong scalability across datasets<br>(WE): Complex sampling adds overhead, and neighborhood sampling may cause information loss, especially in smaller datasets, while ignoring spatial structure                   |
|        | xFraud [20] | (ST): Highly scalable and interpretable, ideal for large transaction networks. The hybrid explanation model improves interpretability, aiding decision-making<br>(WE): Reliance on distributed systems can reduce training efficiency, and performance may degrade with more machines, balancing speed and accuracy |



**Table 2** Comparison of the strengths and weaknesses of transformer models in GNNs

| Aspect               | Model                                            | Strength (ST)/weakness (WE)                                                                                                                                                                                                                                                                                           |
|----------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transformers in GNNs | STAT [23]                                        | (ST): Captures spatial–temporal correlations, enhancing adaptability to data fluctuations, and is versatile across datasets without preset graph structures<br>(WE): High computational demands and performance may decrease with insufficient training data                                                          |
|                      | Finsformer [24]                                  | (ST): Captures complex financial data patterns, improving detection accuracy, and is flexible for various financial threat scenarios<br>(WE): High computational demands and may struggle with noisy or inconsistent data, affecting robustness                                                                       |
|                      | HGT-GNN [25]                                     | (ST): Captures complex relationships in RPT networks, improving fraud detection and adaptability to diverse datasets while avoiding manual feature limitations<br>(WE): High computational demands and challenges in controlling real-world noise, which may impact accuracy                                          |
|                      | DC-GNN [26]                                      | (ST): Addresses camouflage by separating attribute and structure learning, improving detection accuracy in mixed and disguised data environments<br>(WE): High computational demands, especially for complex multi-relational graphs, and scalability issues with larger datasets due to decoupled learning processes |
|                      | STA-GT [27]                                      | (ST): Models local and global transaction data effectively, improving detection accuracy and handling spatial–temporal interactions for long-distance transaction patterns<br>(WE): High computational requirements and complex attention layers may reduce interpretability and real-time applicability              |
|                      | BERT4ETH [28]                                    | (ST): Effectively handles complex transaction patterns in the Ethereum blockchain, improving phishing detection and user account de-anonymization<br>(WE): Requires significant computational resources and lengthy pre-training cycles; de-anonymization tasks need further refinement for optimal results           |
|                      | Distributed knowledge distillation approach [29] | (ST): Handles industry-specific data variations well and provides faster inference with a lightweight student network<br>(WE): Requires training multiple teacher models, increasing complexity, and may struggle with performance on unbalanced datasets                                                             |

**Table 3** Comparison of the strengths and weaknesses of spatial–temporal GNN models

| Aspect                | Model      | Strength (ST)/Weakness (WE)                                                                                                                                                                                                                                                 |
|-----------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spatial–temporal GNNs | STAGN [3]  | (ST): Automatically collects complex temporal and spatial data, ensuring high detection accuracy<br>(WE): Requires significant computational resources and has limited scalability for real-time applications                                                               |
|                       | STGBN [30] | (ST): Efficient incremental learning boosts predictive performance with a unique architecture that reduces training time and improves accuracy<br>(WE): High computational requirements may complicate training, with limited benefits for real-time applications           |
|                       | SIGNN [31] | (ST): Effectively detects gang fraud and individual irregularities by combining spatial and temporal data for improved detection<br>(WE): Complexity in creating heterogeneous graphs may hinder implementation, and high processing requirements could limit practical use |

global information extraction and long-path information aggregation is presented, which adds a global view to local information aggregation. In addition to camouflage detection, the proposed method is designed to more effectively consider long-distance relations between nodes through the attention aggregation operation in the node modeling process.

### 3 Materials and methods

In this section, our proposed method is presented in detail. Firstly, the motivation for the proposed method is illustrated. Next, preliminaries are explained. Finally, we provide an exposition of the proposed method, which consists of five modules. The first module involves the reconstruction of the adjacency matrix of probabilistic relations between fraudulent nodes. The second module focuses on temporal dependency modeling. The third module addresses spatial dependency modeling. The fourth module introduces the transformer layer, and the final module is the prediction layer.

#### 3.1 Motivation of the study

Fraud detection methods using graph neural networks (GNNs) face significant challenges due to the ability of fraudsters to imitate benign behavior, conceal connections, and obscure fraudulent patterns within complex networks. Current GNN-based approaches that rely on local node feature aggregation are often insufficient to address these issues. Fraudsters can mask their malicious activities by blending into the network as seemingly normal nodes, making it difficult to detect them. Additionally, these approaches struggle to capture the timing of fraudulent actions, which is essential for accurately identifying fraud over time. Previous studies [11–16] have largely overlooked these temporal factors, limiting their effectiveness in detecting fraud patterns that unfold across time.

Moreover, prior research [8–10] has not fully leveraged the spatial structure of graphs, which is crucial for identifying hidden, or camouflaged, connections among fraudsters. Many methods fail to detect these deceptive behaviors, reducing their effectiveness. Another significant issue is embedding over-smoothing [32, 33], where the embeddings of nodes become indistinguishable due to hierarchical aggregation, making it harder to differentiate fraudulent from legitimate behaviors. Lastly, earlier works [7] have ignored the intensity of connections between fraudsters, an important factor for accurately identifying fraudulent networks through effective information aggregation.

In response to these problems, our method introduces an approach that aggregates neighboring information for each node by considering both temporal differences of transactions within a specific time slice and dependencies across multiple time slices. We also evaluate the intensity of relationships between nodes to uncover potential fraud linkages. By employing an attention mechanism, our

method adjusts the influence of each neighboring node based on both direct and potential relationships, giving greater weight to nodes that share a higher number of common neighbors with the target node. This attention-based aggregation enhances both local and global information consolidation, improving the ability to detect fraud hidden within complex network structures.

### 3.2 Preliminaries

In this section, we present the foundational concepts and methods that are used in our proposed method.

#### 3.2.1 Defined graph

A graph is formally defined as  $G = \{V, E, X, Y\}$ , where  $V$  and  $E$  are the sets of nodes and edges, respectively. In this context, each node  $v \in V$  represents a user, and each directed edge  $e \in E$  corresponds to a transaction that occurred between two users,  $i$  and  $j$ . The use of directed edges allows us to capture the directionality of transactions, meaning we can distinguish whether user  $i$  initiated a transaction with user  $j$  or vice versa. This directional relationship is crucial for analyzing the flow of information, resources, or actions between users, enabling more accurate modeling of interactions and behaviors within a network.

$X$  is a collection of node feature vectors as  $X = x_1, x, \dots, x_n$ , where each feature vector  $x_i \in \mathbb{R}^d$  represents the attributes of the user  $i$ ,  $d$  is the dimensionality of the feature space, and  $Y$  represents the set of labels for some nodes  $V$ .

#### 3.2.2 Graph autoencoders (GAE)

GAE is an unsupervised model specifically designed for graph representation learning. Its primary objective is to encode the graph structure into a latent space and then reconstruct the original graph, which helps in uncovering hidden or missing links between nodes. In the context of fraud detection, the GAE learns patterns and relationships between nodes (e.g., users, transactions) and reconstructs the adjacency matrix, which represents the probabilistic connections between these nodes. The graph formed by the extracted nodes often resembles a fully connected sub-graph, which GAE processes to identify potential fraudulent relationships.

The functioning of the GAE involves two key phases: encoding and decoding. During the encoding phase, a graph convolutional network (GCN) processes the input graph to create latent node embeddings. These embeddings represent the structural properties and connectivity of the nodes within the graph. For fraudulent node detection, this encoding captures key relationships that help distinguish fraudulent behaviors from normal ones. In the decoding phase, the learned latent embeddings are used to reconstruct the adjacency matrix. The reconstructed matrix provides probabilistic predictions of the relationships between nodes, potentially highlighting hidden links between fraudulent nodes that were not explicitly present in the original data.

The optimization of GAE parameters, such as the weights of the encoder and decoder, is achieved through a gradient-based method, often using the Adam optimizer. The loss function typically used in GAE is a reconstruction loss, which minimizes the difference between the original adjacency matrix and the reconstructed one. This ensures that the model accurately captures the structural dependencies in the graph. Additionally, the optimization process includes regularization techniques, such as dropout, to prevent overfitting and improve the generalization of the model. Hyperparameters like the learning rate, number of layers, and hidden dimensions are fine-tuned using validation data to achieve optimal performance.

The reconstruction module of the adjacency matrix serves as an initialization process for subsequent tasks. The GAE operation is performed separately at the beginning to generate initial embeddings for fraud nodes and a reconstructed adjacency matrix for further use in the model. The primary objective of GAE is to minimize the difference between the original fraud adjacency matrix  $A$  and the reconstructed one  $\hat{A}$ . Hence, the reconstruction loss is typically defined as:

$$L_{\text{reconstruction}} = -\frac{1}{N^2} \sum_{i,j} [A_{ij} \log(\hat{A}_{ij}) + (1 - A_{ij}) \log(1 - \hat{A}_{ij})] \quad (1)$$

This formulation represents the cross-entropy loss, capturing the probabilistic relationships between nodes. To prevent overfitting and improve generalization, regularization techniques such as dropout and  $L_2$ -norm are applied. The regularization loss is defined as:

$$L_{\text{regularization}} = \lambda \|\theta\|_2^2 \quad (2)$$

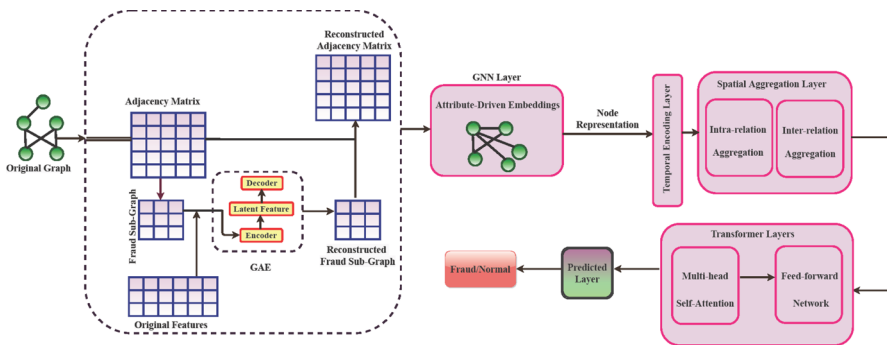
where  $\lambda$  is a hyperparameter controlling the strength of regularization. The final loss function combines the reconstruction loss and the regularization term:

$$L_{\text{total}} = L_{\text{reconstruction}} + L_{\text{regularization}} \quad (3)$$

The total loss function  $L_{\text{total}}$  is minimized using gradient-based optimization methods, such as the Adam optimizer. To achieve optimal performance, hyperparameters like the learning rate, number of GCN layers, and dimensions of the latent space are fine-tuned using validation data.

### 3.3 Proposed method

In this subsection, the proposed method is introduced and explained in detail. The proposed method consists of two approaches called proposed-method-WN and proposed-method-WN-CN using five modules, where WN and CN represent weighted neighbors and common neighbors, respectively. The first module focuses on reconstructing the adjacency matrix of probabilistic relations among fraudulent nodes, utilizing graph autoencoders (GAE) and cosine similarity to identify potential links between these nodes. The second module involves temporal dependency modeling, employing a temporal encoding approach to effectively capture temporal relationships. The third module is dedicated to spatial dependency modeling, aggregating



**Fig. 1** Proposed method

neighboring information for each node at each time slice as well as across different time slices. The fourth module, the transformer layer, aggregates global information derived from the combined results of the temporal- and spatial-aware modeling for each node. Finally, the prediction layer module is responsible for classifying whether a node is fraudulent or not. Figure 1 illustrates the proposed method.

The following subsections provide a detailed explanation of each component of the proposed framework, and the pseudocode of the proposed framework is presented in Algorithm 1. Section 3.3.1 focuses on the adjacency matrix reconstruction module, which is designed to uncover potential relationships between fraud nodes, as well as the GNN layer that generates attribute-driven embeddings for all nodes. The output of this component is node embeddings, which are forwarded to the temporal encoding layer, described in Sect. 3.3.2. The output of the temporal encoding layer—updated node embeddings—is then passed to the spatial aggregation layer, covered in Sect. 3.3.3. In this layer, after performing inter-relation and intra-relation aggregation, the node embeddings are forwarded to the transformer layer, detailed in Sect. 3.3.4. Finally, the prediction layer, described in Sect. 3.3.5, performs node classification (fraud/normal).

## Algorithm 1 Steps of the proposed method

**Input:** Initial adjacency matrix  $A$ , feature matrix  $X$

**Output:** Predicted labels for nodes (fraud or normal)

**Function** Proposed Method:

### Step 1: Reconstruct Adjacency Matrix

**Input:** Initial adjacency matrix  $A$

**Output:** Reconstructed adjacency matrix  $A_{\text{reconstructed}}$

1. Extract the fraud subgraph  $G_{\text{fraud}}$  from  $A$ .
2. Generate node embeddings for  $G_{\text{fraud}}$  using Graph Auto-Encoder (GAE).
3. Discover hidden connections among fraud nodes and update  $G_{\text{fraud}}$ .
4. Replace  $G_{\text{fraud}}$  in  $A$  with the updated  $G_{\text{fraud}}$  to form  $A_{\text{reconstructed}}$ .

### Step 2: Node Embedding with GNN Layers

**Input:** Reconstructed adjacency matrix  $A_{\text{reconstructed}}$ , feature matrix  $X$

**Output:** Node embeddings  $H^{(1)}, H^{(2)}, \dots, H^{(L)}$

**for** each layer in GNN **do**

Aggregate neighborhood information for each node.

Update node embeddings using GNN layer operation.

**end**

### Step 3: Temporal Embedding Layer

**Input:** Node embeddings  $H^{(1)}, H^{(2)}, \dots, H^{(L)}$

**Output:** Temporal embeddings  $H_{\text{temporal}}$

**for** each node **do**

Encode temporal dependencies from node embeddings using Equations (2) and (3).

**end**

### Step 4: Spatial Aggregation Layer

**Input:** Temporal embeddings  $H_{\text{temporal}}$

**Output:** Aggregated spatial-temporal embeddings  $H_{\text{spatial}}$

Perform inter-relation aggregation to capture global relationships using Equation (5).

Perform intra-relation aggregation to refine local relationships using Equation (4).

### Step 5: Multi-Head Transformer Layer

**Input:** Aggregated spatial-temporal embeddings  $H_{\text{spatial}}$

**Output:** Final node embeddings  $H_{\text{final}}$

Apply a multi-head attention mechanism to capture long-range dependencies.

Generate enhanced embeddings using the transformer layer.

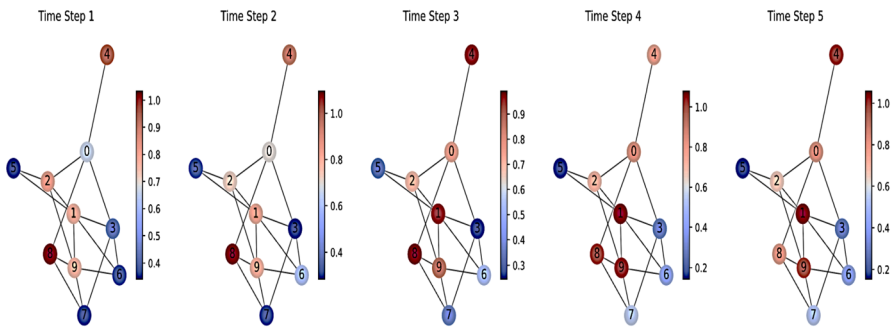
### Step 6: Prediction Layer

**Input:** Final node embeddings  $H_{\text{final}}$

**Output:** Predicted labels for nodes (fraud or normal)

Pass embeddings through a multi-layer perceptron (MLP) classifier.

Predict probabilities for fraud detection.



**Fig. 2** Capturing the temporal pattern of the GNN layer, with darker colors indicating nodes with stronger temporal activity at each time stamp

Each module of the proposed method is described below.

### 3.3.1 Reconstruction module of the adjacency matrix for probabilistic relations between fraudulent nodes

The motivation for incorporating a reconstruction module lies in addressing the confusion of features that arise from concealing the connections between fraudulent nodes. A portion of the indirect connections among neighboring fraudulent nodes is derived through a higher-order adjacency matrix. This module establishes connections between fraudulent nodes using long-distance meta-paths, which capture the complex interrelations that exist beyond immediate neighbors.

The first step in obtaining the probability of hidden connections between fraudulent nodes is to construct the higher-order adjacency matrix. This matrix represents long-path relations ( $k$ -order) between nodes, effectively capturing indirect relationships that may indicate fraudulent behavior. From this higher-order adjacency matrix, we extract a specific matrix of fraudulent nodes, referred to as the Fraud-AdjMat. This matrix has equal rows and columns corresponding to the number of fraudulent nodes, facilitating focused analysis of their interconnections.

The rationale for employing a higher-order adjacency matrix lies in its ability to unveil deeper relational patterns that are not evident in first-order connections. Fraudulent actors often manipulate relationships with legitimate users to obfuscate their actions. By leveraging higher-order connections, we can identify potential collusions or coordinated behaviors among fraudulent nodes that would remain hidden in a simpler graph representation.

To quantify the probability of hidden connections between the fraudulent nodes present in the Fraud-AdjMat matrix, we utilize the GAE approach. This technique allows the representation of node connections through latent embeddings. Specifically, we create a new matrix by multiplying the Fraud-AdjMat matrix with the feature matrix of the fraudulent nodes, which is then fed into the GAE. This process generates latent representations that effectively capture the underlying relationships among the fraudulent nodes.



Moreover, cosine similarity is calculated between latent representations of fraudulent nodes, and zero entries in the original adjacency matrix are updated if the similarity exceeds 0.5, reconstructing the matrix to reflect potential connections better.

In the next phase, we focus on extracting temporal and spatial features between the nodes. The feature matrix for the nodes is calculated using the GNN approach. GNNs are adept at learning representations (embeddings) for nodes in a structured graph. The adjacency matrix plays a vital role in GNNs as it encodes the graph's structure, representing connections between nodes. The convolution operations within GNNs involve applying filters to each node and its neighbors, aggregating information from neighboring nodes using learnable parameters, and subsequently updating the features of each node. These operations empower GNNs to extract local information and generalize it across the graph.

As shown in Fig. 2, the GNN layer captures discrete temporal patterns by adjusting node encodings to reflect changes in interactions over time. Darker color intensities represent stronger temporal activity, indicating nodes with higher temporal influence or activity at each specific time step. This visualization provides an intuitive understanding of the temporal and spatial dependencies within the graph, as nodes more involved in temporal interactions are easily identifiable based on color shifts.

Therefore, the visualization confirms the GNN's ability to capture and represent temporal dependencies crucial for fraud detection, with color intensity effectively highlighting the nodes' temporal significance and interaction patterns across the graph.

In our method, the attribute-driven embedding layer is employed to facilitate feature similarity learning without relying solely on the graph's topological information. For a given node  $v_i$ , the primary layer embedding is represented as follows:

$$h_{v_i}^o = \sigma(x_i W_1) \quad (4)$$

In this equation,  $\sigma$  denotes a nonlinear activation function,  $x_i \in \mathbb{R}^s$  represents the original features of node  $v_i$ , and  $W_1 \in \mathbb{R}^{s \times d}$  is a learnable weight matrix, where  $s$  represents the dimensionality of the input feature space, and  $d$  denotes the dimensionality of the embedding space.

Regarding Eq. (4),  $W_1$  is essential for transforming the original feature vector  $x_i$  of node  $v_i$  into its embedding representation  $h_{v_i}^o$  and facilitates linear transformations of the input feature vector, allowing the model to learn relevant representations during training. The nonlinear activation function  $\sigma$  further enhances the expressiveness of the learned embeddings by introducing nonlinearity into the transformation. Consequently,  $W_1$  is a critical component of the attribute-driven embedding layer, enabling meaningful embeddings to be learned from the original node features without dependence on the underlying graph structure.

In summary, once the probabilistic reconstructed adjacency matrix of size  $n \times n$  is obtained, it is provided to the GNN. Utilizing the neighboring information of the nodes, the feature matrix for each node is generated, resulting in an  $n \times (\text{number of features})$  matrix, where  $n$  corresponds to the total number of nodes in the graph.

### 3.3.2 Temporal encoding layer

In the fraud detection process, separate graphs for each time slice cannot achieve the structural constraints required to capture the interactions between the target node and nodes in different time slices. Our motivation for introducing the temporal encoding layer is to effectively capture temporal information and maximize the utilization of structural information. This layer allows nodes to learn a hidden temporal representation, bridging the gaps between time slices and enhancing the overall detection process. The time encoding for a target node  $v_i$  at time  $t(v_i)$  can be written as follows [27]:

$$\begin{aligned}\text{Base}(t(v_i), 2i) &= \sin\left(\frac{t_{v_i}}{10000^{\frac{2i}{d}}}\right), \\ \text{Base}(t(v_i), 2i+1) &= \cos\left(\frac{t_{v_i}}{10000^{\frac{2i+1}{d}}}\right), \\ \text{TE}(t_{v_i}) &= T - \text{Linear}(\text{Base}(t(v_i)))\end{aligned}\quad (5)$$

where  $T - \text{Linear}$  is an adjustable linear projection. This method utilizes the sine and cosine functions to create a periodic representation of time, which captures essential properties such as trends, smoothness, and phase shifts. These trigonometric functions are inherently periodic and can effectively model the cyclical nature of various phenomena, such as user activity patterns, transaction timings, and seasonal variations.

By encoding time in this manner, we allow for the representation of temporal relationships between nodes, enabling the model to distinguish not only when an event occurs but also how it relates to previous and subsequent events. This temporal representation is critical because it serves as an initial embedding for spatial learning, helping to improve the model's ability to recognize and react to patterns of fraud that may occur over time.

The relative temporal dependency between nodes  $v_i$  and  $v_j$  can therefore be modeled effectively. The hidden representation of node  $v_i$  can be updated as below by including time encoding:

$$h_{v_i}^{o,t} = h_{v_i}^o + \text{TE}(t_{v_i}) \quad (6)$$

In this equation, the hidden representation  $h_{v_i}^{o,t}$  incorporates the temporal encoding.

$\text{TE}(t_{v_i})$ , thereby enriching the node's feature set with time-sensitive information. This enriched representation not only enhances the model's ability to detect fraudulent patterns but also allows for more nuanced interactions among nodes that occur at different times.

By providing a robust framework for temporal dependency modeling, we aim to equip our model with the necessary tools to effectively learn from complex, time-dependent data, ultimately improving fraud detection outcomes.

This process leverages two key functions,  $\text{Base}(\cdot)$  and  $\text{TE}(\cdot)$ , which are critical for encoding temporal information.

$\text{Base}(t(v_i), 2i)$  serves as a positional encoding for node timestamps, using sinusoidal functions to encode temporal information continuously, similar to transformer models. This encoding captures both short and long time intervals, introducing periodicity to better reflect patterns in fraudulent behavior.  $\text{TE}(t(v_i))$ , i.e., temporal encoding, further transforms this base temporal information into latent representations through a linear projection, making it suitable for use in graph-based operations.

The  $T - \text{Linear}(\cdot)$  function projects temporal encodings into a hidden space, allowing the model to combine both temporal and structural information. This helps the GNN learn how time influences interactions between nodes. For example, two transactions that occur closer in time may exhibit stronger relationships than those further apart. The temporal encoding is crucial for fraud detection as it highlights patterns where fraudsters often operate within specific time windows.

In summary, these temporal encodings allow the model to incorporate time into node interactions, providing a more comprehensive representation of fraud dynamics. This method improves the GNN's ability to identify fraud by taking into account the evolution of relationships over time.

### 3.3.3 Spatial aggregation layer

Fraudsters tend to share similar features to avoid detection in the transaction network. A transaction graph  $G$  encodes the relations between transactions. The GNN method is used to model spatial dependency, and the information between layers and within layers is aggregated. After that, by adding intermediate layer embeddings, a comprehensive representation is obtained that includes both spatial and temporal patterns.

**3.3.3.1 Intra-layer aggregation** According to node  $v_i$  and its neighboring node  $v_j$  in layer  $l$ , neighboring information is trained. The graph convolution function is shown below [27].

$$\begin{aligned} h_{i,r}^{l,t} &= \text{COMBINE}\left(\text{AGGR}\left(W_{v_i,v_j}\left\{h_{v_j}^{l-1,t'}, v_j \in N_r(v_i)\right\}\right)\right), \\ \text{AGGR}\left(\left\{\alpha\left(h_{v_j}^{l-1,t} - h_{v_j}^{l-1,t'}\right)\right\}, v_j \in N_r(v_i)\right) & \\ \alpha &= \left|N_r(v_i) \cap N_r(v_j)\right| \end{aligned} \quad (7)$$

where  $h_{v_j}^{l-1,t}$  represents the  $l$ -th layer embedded with the temporal information of node  $v_i$ ,  $l \in \{1, 2, \dots, L\}$ ,  $t$  represents the timestamp of node  $v_i$ ,  $t'$  represents the timestamp of node  $v_j$ , and  $N(v_i)$  is the set of neighbors of node  $v_i$ . Each node  $v_j$  is one of the neighbors of node  $v_i$  if its  $W_{v_i,v_j}$  value in the reconstructed adjacency matrix is nonzero, i.e., 1 or a float value.

In the above formula,  $W_{v_i, v_j}$  represents the probabilistic relation of nodes  $v_i$  and  $v_j$  in the reconstructed adjacency matrix, which can be zero, one, or a float number. In other words, weighted summation is used to combine the embedding of nodes  $h_{v_j}^{l-1, n}$ .

This weighted integration helps to effectively extract spatial information for fraudulent nodes in such a way that for a fraudulent node, the embedding information of its long-distance fraudulent neighbors is combined with an appropriate weighting factor, and accurate spatial information is produced.

In the node embedding extraction process, parameter  $\alpha$  calculates the number of common nodes. Therefore, if  $v_i$  and  $v_j$  have more common neighbors, the feature vectors of  $v_i$  and  $v_j$  will be closer, and their differences will be smaller. Using parameter  $\alpha$  leads to better extraction of spatial information for benign nodes in such a way that two nodes that have more neighbors in common have more similar node embeddings than two nodes that are far from each other, and this results in faster and better convergence of benign node embeddings.

In spatial dependency modeling, both parameters  $W_{v_i, v_j}$  and  $\alpha$  are used for intra-layer aggregation in the proposed-method-WN-CN, while  $W_{v_i, v_j}$  is utilized in the proposed-method-WN.

The following details provide further explanations of the two methods, method-WN (weighted neighbors) and method-WN-CN (weighted neighbors with common neighbors). The motivation behind these methods is to enhance fraud detection by effectively aggregating information from neighboring nodes and capturing patterns indicative of fraudulent behavior. While both methods utilize the same underlying five modules, they differ primarily in how they handle neighbor information for fraud detection.

Method-WN focuses solely on the weighted neighbors of each node. Its motivation lies in emphasizing the strength of relationships based on probabilistic connections, where weights in the adjacency matrix are determined by the intensity of the relationship between nodes. This helps in aggregating information based on the direct influences of neighboring nodes.

In contrast, method-WN-CN incorporates not only the weighted neighbors but also emphasizes common neighbors among the nodes. The motivation for this method is to aggregate information based on both the weighted relationships and the number of shared neighbors. This dual consideration enables the method to better account for the collaborative behaviors of fraudulent nodes, providing a more robust analysis by leveraging both direct connections and shared relationships.

Method-WN primarily relies on the probabilistic relations defined in the reconstructed adjacency matrix, where connections are inferred based on the strength of direct relationships. Method-WN-CN enhances this by integrating the common neighbor approach, which enables the model to utilize shared connections to strengthen the aggregation process. This allows the model to extract richer contextual information from the network structure, thereby improving its ability to detect fraud.

In terms of performance and robustness, method-WN may be more effective in environments where direct connections are strong but may struggle with detecting more subtle patterns of fraud that involve common relationships. On the other

hand, method-WN-CN, by taking common neighbors into account, is likely to perform better in scenarios where fraudsters attempt to hide within larger groups or use shared connections to disguise their activities.

Overall, the motivation behind both methods is to enhance fraud detection capabilities through innovative neighbor aggregation strategies. While method-WN focuses on weighted relationships, method-WN-CN offers a more comprehensive approach by leveraging both weighted and common neighbor information, potentially leading to improved fraud detection.

**3.3.3.2 Inter-layer aggregation** The motivation for inter-layer aggregation lies in the need to combine both local and global information effectively to enhance spatial–temporal information modeling. The primary layers of the network mainly capture local information, while the later layers represent global information. To address this, the intermediate embeddings obtained from the GNN layers are connected:

$$h_{v_i}^t = \text{COMBINE}\left(h_{v_i}^{1,t}, h_{v_i}^{2,t}, \dots, h_{v_i}^{L,t}\right) \quad (8)$$

This approach ensures that the model leverages both granular details and overarching patterns, providing a comprehensive representation of spatial–temporal dependencies.

### 3.3.4 Transformer layer

The motivation for employing the transformer layer in this work lies in its ability to capture complex interactions among node embeddings, enhancing their representation for downstream tasks. This choice ensures that the model can effectively process and leverage both temporal and spatial dependencies in the data. The integration of transformers and multi-head attention mechanisms has recently been applied in various domains, such as natural language processing and computer vision. It is now making headway in graph neural networks (GNNs) for fraud detection [7]. This innovative application leverages the unique strengths of transformers and multi-head attention to address challenges specific to fraud detection, enhancing the GNN's ability to identify complex fraud patterns.

Fraud detection with GNNs faces challenges such as node camouflage, where fraudsters mimic legitimate behavior, over-smoothing during neighborhood aggregation, and difficulty in capturing spatial–temporal dependencies. The motivation behind using multi-head attention is its ability to address these issues by enabling GNNs to focus on diverse relational patterns, highlight key interactions, and isolate hidden fraudulent activities. Similarly, transformers enhance GNNs by capturing global structural patterns and long-range dependencies, allowing for the detection of subtle fraud schemes across distributed transactions. Together, these mechanisms improve fraud detection capabilities, but their effectiveness relies on careful attention mechanism design, hyperparameter tuning, and empirical validation.

The transformer layer begins by using the aggregated spatial–temporal embeddings  $H_{\text{spatial}} \in \mathbb{R}^{N \times d}$ , which represent the node embeddings. Here,  $N$  denotes the

**Table 4** Three samples of the real-world dataset

| Feature       | Sample 1             | Sample 2             | Sample 3             |
|---------------|----------------------|----------------------|----------------------|
| Primary_ID    | 992e7ca6cc0d738d85b2 | 737b3293b0d5ade9ad90 | 737b3293b0d5ade9ad90 |
| TRNS_Type     | 10                   | 10                   | 10                   |
| Date          | 7/20/2022            | 7/6/2022             | 7/8/2022             |
| Time          | 20:00                | 20:00                | 20:00                |
| Amount        | 872,000              | 1,700,000            | 3,400,000            |
| Terminal_Type | 7                    | 1                    | 1                    |
| Terminal_ID   | 13ae6270758cdba26ec1 | f52c0e50a8eb31e3965c | f52c0e50a8eb31e3965c |
| Second_ID     | 8457468be95f8a197756 | 9d53d2f9c5fe1fc7f260 | 9d53d2f9c5fe1fc7f260 |
| Class (fraud) | N                    | Y                    | N                    |

**Table 5** Specifications of the real-world dataset

| Dataset    | #Transactions | #Samples  | #Normal | #Fraudulent | #Features |
|------------|---------------|-----------|---------|-------------|-----------|
| Real-world | 4,064,194     | 1,457,760 | 20,444  | 6,765       | 8         |

number of nodes, and  $d$  represents the dimension of each node embedding. This input matrix is passed through a multi-head attention mechanism to extract both local and global dependencies among nodes. The multi-head attention mechanism first computes the scaled dot-product attention for each head, mathematically expressed as  $Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$ . In this equation,  $Q$ ,  $K$ , and  $V$  are the query, key, and value matrices, respectively, which are derived from  $H_{spatial}$  using learnable weight matrices  $W_Q$ ,  $W_K$ , and  $W_V$ . The outputs from all attention heads are then concatenated and projected back to the original dimension using another learnable matrix  $W_O$ .

The output of the transformer layer is an updated embedding matrix as final node embeddings  $H_{final}$ , where each node's representation incorporates both local and global information. This matrix contains enriched embeddings for all nodes, capturing intricate spatial-temporal dependencies and long-range interactions across the graph. These refined embeddings are better suited for downstream tasks, such as fraud detection, as they leverage the multi-head attention mechanism to emphasize critical relationships and mitigate challenges like over-smoothing and node camouflage.

### 3.3.5 Prediction layer

The MLP classifier is utilized to classify nodes, i.e., identify fraudulent transactions. The optimization of this process is performed by the cross-entropy loss function, where  $y_v$  indicates the real label of node  $v$ .

**Table 6** Statistics of graph datasets for YelpChi and amazon datasets

| Dataset | #Nodes (Fraud %) | #Edges    | Relations         | #Relations                  |
|---------|------------------|-----------|-------------------|-----------------------------|
| YelpChi | 45,954 (14.5%)   | 3,846,979 | R-U-R R-S-R R-T-R | 49,315 3,402,743 573,616    |
| Amazon  | 11,944 (9.5%)    | 4,398,392 | U-P-U U-S-U U-V-U | 175,608 3,566,479 1,036,737 |

$$L = - \sum_{v \in V} y_v \log P_v + (1 - y_v) \log (1 - P_v), \quad (9)$$

$$P_v = \sigma(MLP(z_v))$$

## 4 Experiments

In this section, we conduct fraud detection experiments on real-world and benchmark datasets. In this paper, codes are simulated via Python software. The system used for experimental evaluation is a Geforce RTX 2080 Ti with a CPU of 8 cores, 24 GB of RAM, and 150 GB of HDD.

### 4.1 Real-world dataset

The real-world dataset consists of 4,064,194 transactions by Iranian bank customers related to the period of three months in 2022. The credit card transactions are denoted by Primary\_ID and Second\_ID as the source and destination of a transaction, with associated features such as transaction type (TRNS\_Type), date, time, amount, terminal type, and Terminal\_ID, including 1,457,760 samples (1,424,061 unique Primary\_IDs and 33,699 unique Second\_IDs). The dataset is representative of a variety of credit card activities, including both normal and potentially fraudulent transactions, and provides a comprehensive overview of credit card transactions, making it a valuable resource for developing and testing fraud detection models within the financial domain. This dataset consists of seven terminal types, such as internet, mobile, phone, POS, branch, kiosk, and ATM, which are identified by numbers one to seven; three transaction types, including card-to-card, purchase, and payment, denoted by 10, 20, and 30, respectively, and classes that consist of fraud and benign, which are indicated by Y representing fraud and N representing benign or not fraud. Table 4 shows three samples of this dataset. A summary of the dataset specifications is given in Table 5.

### 4.2 Benchmark datasets

We utilized two benchmark datasets in this study: YelpChi and Amazon. The YelpChi dataset is a multi-relational graph constructed from the Yelp spam review dataset, designed for evaluating models in node classification, fraud detection,

and anomaly detection. It focuses on distinguishing between spam and legitimate reviews in a binary classification task. In this dataset, reviews are represented as nodes, and three specific types of relationships are established: R-U-R connects reviews posted by the same user; R-S-R connects reviews associated with the same product that has the same star rating (ranging from 1 to 5 stars); and R-T-R connects two reviews for the same product that were posted within the same month. While these relationships explicitly define certain temporal dependencies (e.g., reviews posted in the same month), the dataset also implicitly incorporates temporal information within its metadata. Specifically, timestamps of reviews are available as part of the dataset, and these timestamps can be utilized to uncover and model additional temporal patterns. This implicit temporal aspect enables our approach to enhance its understanding of temporal dependencies. After preprocessing, the YelpChi dataset includes 45,954 reviews (14.5% fraud) and provides 32 handcrafted primary features for each node [7].

The Amazon-Fraud dataset is a multi-relational graph constructed from the Amazon review dataset, specifically focusing on product reviews within the Musical Instruments category. In this dataset, users are classified as benign entities if they have more than 80% helpful votes, while those with less than 20% helpful votes are labeled as fraudulent entities. This creates a binary classification task for detecting fraudulent users. The dataset includes 25 handcrafted features derived from previous research, which serve as the raw node features for the Amazon-Fraud graph. In this graph, users are represented as nodes, with three specific types of relationships established: U-P-U connects users who have reviewed at least one identical product; U-S-U connects users who have assigned at least one identical star rating within one week; and U-V-U connects users with the top 5% of mutual review text similarities, measured by TF-IDF, among all users. Although these relationships provide explicit insights into user behavior, the dataset also contains implicit temporal information through timestamps of reviews. These timestamps allow us to infer temporal patterns and further augment the explicit relationships. The Amazon-Fraud dataset, after preprocessing, includes product reviews labeled into two categories: normal and fraudulent. It provides 25 handcrafted primary features to support effective modeling [7]. The statistics of graph datasets for YelpChi and Amazon datasets are shown in Table 6.

### 4.3 Baseline methods

To enhance the comparative understanding of the proposed methods, we provided brief descriptions of the baseline methods used in the evaluation. These methods are GNN-based approaches that focus on various aspects of fraud detection:

CIES-GNN (camouflage identification and enhanced semantics aggregation graph neural network) is a specialized method for fraud detection that employs collaborative inference to improve model performance by leveraging both node features and structural information.



GCN (Graph Convolutional Network) [7]: A foundational GNN model that employs spectral convolution to learn node representations by aggregating features from neighboring nodes. GCN is effective in capturing local structures within graph data.

GAT (graph attention network) [8]: This spatial GNN model incorporates an attention mechanism, allowing it to weigh the importance of different neighbors during the aggregation process. This enables the model to focus on the most influential nodes for learning representations.

GraphSAGE [27]: Designed to address memory explosion issues, GraphSAGE utilizes neighborhood sampling to efficiently aggregate information from a fixed-size neighborhood, making it suitable for large graphs while supporting inductive learning.

DR-GCN (denoising regularized graph convolutional network) [26]: This GNN variant introduces a denoising mechanism to handle noisy data effectively, improving the quality of node representations and enhancing performance in fraud detection scenarios.

GraphConsis [3]: A heterogeneous graph model aimed at optimizing balanced sampling through the measurement of neighbor node distances. It is primarily applied in risk control scenarios to ensure robust detection of fraudulent activities.

CARE-GNN [34]: An end-to-end model focused on addressing the challenges of fraudsters employing node camouflage techniques. This model enhances the detection of fraudulent nodes by specifically targeting deceptive behaviors.

PC-GNN (prototypical class graph neural network) [10]: This fraud detection model tackles sample imbalance issues through undersampling and utilizes node distance measurements to effectively resample adjacent nodes, improving classification performance.

GPRGNN (graph propagation and regression GNN) [28]: This model optimizes the extraction of topological information and node features using a graph propagation regression architecture that learns adaptive weights for better representation.

FAGCN (feature augmented graph convolutional network) [29]: A spectral domain model that proposes extracting convolution kernels for high-frequency and low-frequency information separately, enhancing the representation capabilities of the network.

Rio-GNN (reinforced, recursive, and scalable multi-relational graph neural network) [9]: This architecture is guided by neighborhood selection, focusing on reinforcement learning to improve scalability and adaptability in fraud detection tasks.

AO-GNN (attention-based outlier detection GNN) [30]: This model determines the connectivity of each node's neighbors and maximizes the area under the curve (AUC) through reinforcement learning, enhancing its sensitivity to detecting outliers.

H2-FDetector [13]: A specialized fraud detection model that identifies node camouflage by classifying connections into homophilic and heterophilic interactions, enabling a more nuanced understanding of fraudulent behavior.

**Table 7** Performance of different methods on the real-world dataset

| Metrics  | GCN    | GAT    | CIES-GNN | Proposed-method-WN | Proposed-method-WN-CN |
|----------|--------|--------|----------|--------------------|-----------------------|
| AUC      | 0.6797 | 0.6796 | 0.8658   | 0.8747             | 0.8762                |
| F1-macro | 0.6377 | 0.5709 | 0.7100   | 0.7129             | 0.7170                |
| Gmean    | 0.4951 | 0.1935 | 0.7856   | 0.7876             | 0.7894                |

By comparing the proposed methods (proposed-method-WN and proposed-method-WN-CN) against these established GNN-based approaches, we aim to demonstrate their effectiveness in capturing the intricate relationships necessary for accurate fraud detection in the YelpChi and Amazon datasets. Each baseline method is referenced to provide a comprehensive understanding of their respective contributions to the field of fraud detection.

#### 4.4 Evaluation metrics

Since graph fraud detection is defined as a classification problem, the proposed method is evaluated by AUC, F1-macro, and Gmean criteria [7].

##### 4.4.1 AUC

The area under the ROC Curve (AUC) criterion is calculated based on the prediction probability ranking of all samples, which removes the effect of data imbalance in the detection process.

$$AUC = \frac{1}{2} \sum_{i=1}^{m=1} (TPR_{i+1} - TPR) (FPR_{i+1} - FPR_i) \quad (10)$$

where TP, FN, FP, TN, TPR ( $TPR = TP / (TP + FN)$ ), and FPR ( $FPR = TN / (TN + FP)$ ) represent true positive, false negative, false positive, true negative, true-positive rate, and false-positive rate, respectively.

##### 4.4.2 F1-macro

The F1-macro criterion is the harmonic mean of recall and precision. The recall measure shows the efficiency of the classifier in identifying real fraudulent transactions, and the precision measure indicates the reliability of the classifier output.

$$F1 - macro = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (11)$$

**Table 8** Performance of different methods on YelpChi and amazon datasets

| Method    | Dataset metrics       | YelpChi |          |        | Amazon |          |        |
|-----------|-----------------------|---------|----------|--------|--------|----------|--------|
|           |                       | AUC     | F1-macro | Gmean  | AUC    | F1-macro | Gmean  |
| Baselines | GCN                   | 0.5992  | 0.5622   | 0.4365 | 0.8374 | 0.6417   | 0.5723 |
|           | GAT                   | 0.5825  | 0.4893   | 0.1659 | 0.8142 | 0.6554   | 0.6678 |
|           | GraphSAGE             | 0.5583  | 0.4517   | 0.2589 | 0.7591 | 0.6525   | 0.5951 |
|           | DR-GCN                | 0.5968  | 0.5673   | 0.4038 | 0.8297 | 0.6518   | 0.7972 |
|           | GPRGNN                | 0.7967  | 0.5967   | 0.7167 | 0.9462 | 0.7355   | 0.8711 |
|           | FAGCN                 | 0.7734  | 0.5658   | 0.6845 | 0.9497 | 0.8303   | 0.8892 |
|           | GraphConsis           | 0.6995  | 0.5982   | 0.5857 | 0.8755 | 0.7559   | 0.7684 |
|           | CARE-GNN              | 0.7668  | 0.6443   | 0.6791 | 0.9073 | 0.8992   | 0.8967 |
|           | PC-GNN                | 0.8226  | 0.6556   | 0.7395 | 0.9591 | 0.8945   | 0.9102 |
|           | Rio-GNN               | 0.8353  | 0.6581   | 0.7383 | 0.9623 | 0.8973   | 0.9147 |
|           | AO-GNN                | 0.8812  | 0.7056   | 0.8123 | 0.9653 | 0.8924   | 0.9097 |
|           | H2-FDetector          | 0.8879  | 0.6967   | 0.8162 | 0.9691 | 0.8384   | 0.9211 |
|           | CIES-GNN              | 0.8884  | 0.7285   | 0.8061 | 0.9747 | 0.875    | 0.9251 |
| Ours      | Proposed-method-WN    | 0.8952  | 0.7325   | 0.8065 | 0.9705 | 0.8652   | 0.922  |
|           | Proposed-method-WN-CN | 0.9001  | 0.7336   | 0.8105 | 0.9775 | 0.8852   | 0.9302 |

#### 4.4.3 Gmean

The geometric mean (Gmean) combines the recall and precision of a classifier to evaluate the performance of a classification model on an unbalanced dataset and is defined as:

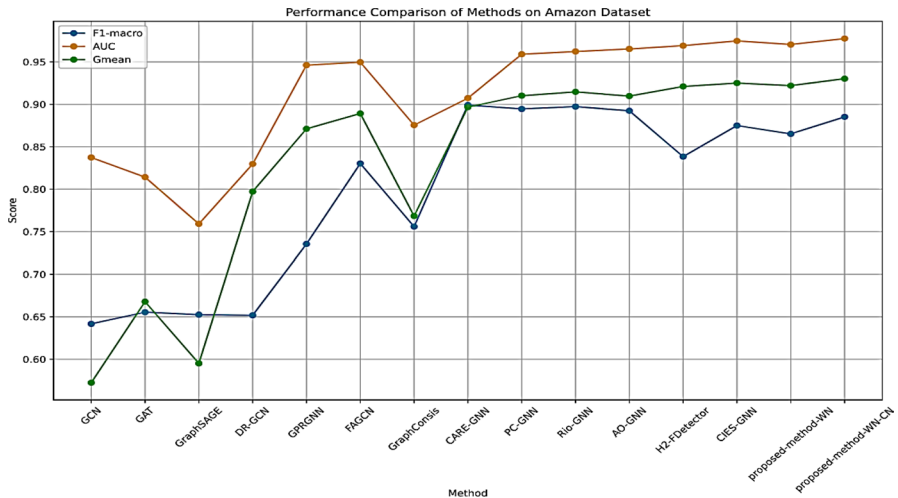
$$G_{\text{mean}} = \sqrt{\text{TPR} \times \text{TNR}} \quad (12)$$

#### 4.5 Experiments on the real-world dataset

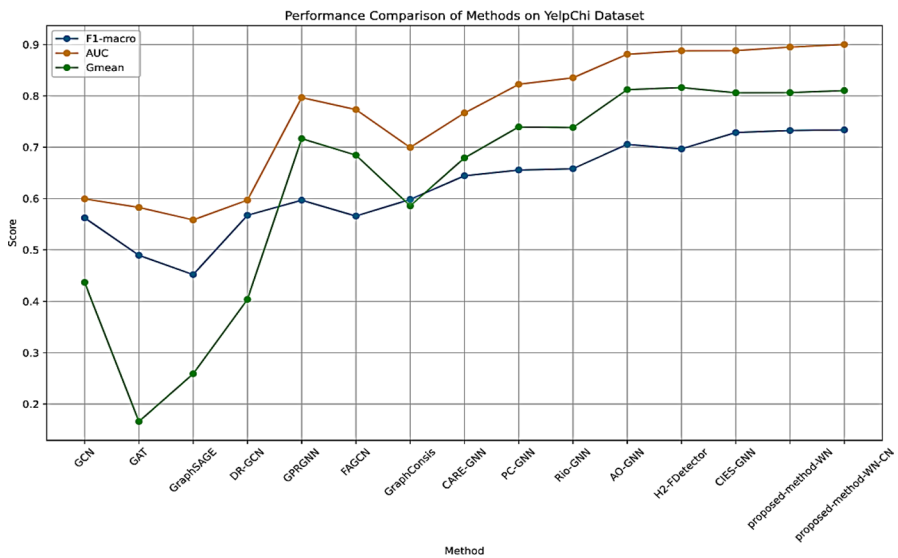
In all experiments, 10 iterations were performed, and the average values were considered the final result. The proportions of training, testing, and validation are 40%, 40%, and 20%, respectively. The dropout rate and the positive sample oversampling rate are both equal to 0.5, and the number of training epochs is 150. The evaluation results for the real-world dataset are shown in Table 7.

Table 7 presents the performance metrics of various methods for fraud detection on the real-world dataset, which comprises a significant number of transactions. The methods evaluated include GCN, GAT, CIES-GNN, and two proposed methods denoted as proposed-method-WN and proposed-method-WN-CN. The evaluation metrics considered are AUC, F1-macro, and Gmean.

Among the methods, CIES-GNN exhibits notable performance, achieving an F1-macro score of 0.710 and an AUC of 0.866. This method demonstrates a balanced trade-off between precision and recall, as indicated by its relatively high

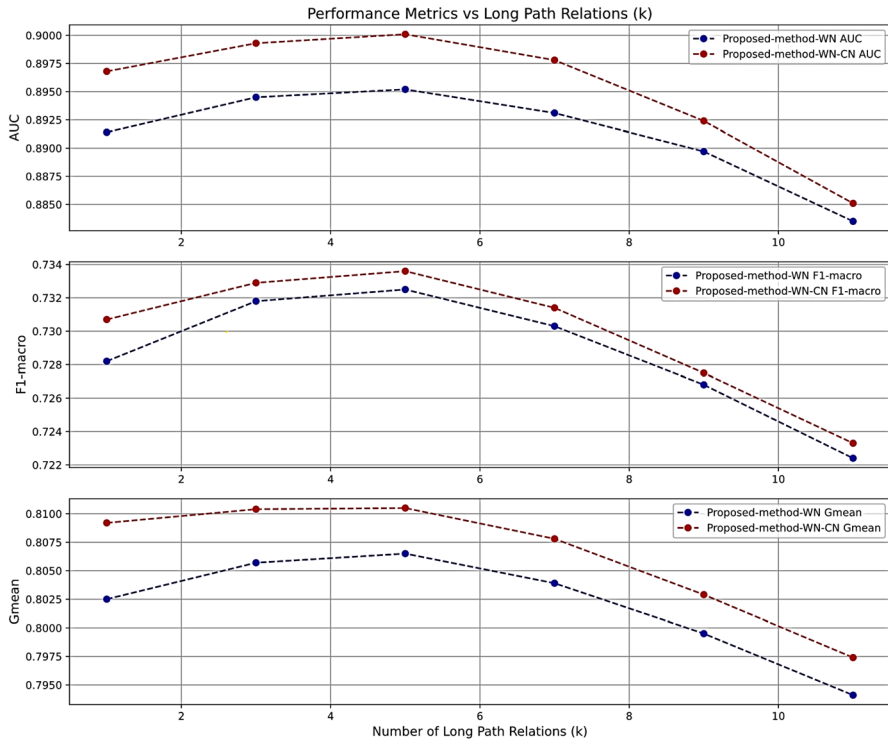


**Fig. 3** Comparison of the methods based on evaluation criteria for the Amazon dataset



**Fig. 4** Comparison of the methods based on evaluation criteria for the YelpChi dataset

F1-macro score. However, it is surpassed by the proposed methods, which incorporate additional enhancements. Both proposed-method-WN and proposed-method-WN-CN yield superior performance, with F1-macro scores of 0.713 and 0.717, respectively, and AUC values above 0.874. These proposed methods leverage weighted neighbors (WN) and common neighbors (CN), which likely contribute to their improved discriminatory power.



**Fig. 5** Evaluation of hyperparameter sensitivity in long-path relations for fraud detection

Interestingly, traditional GNN architectures like GCN and GAT exhibit mixed performance. While GCN achieves a respectable F1-macro score of 0.638, GAT falls short with a score of 0.571. This suggests that the choice of graph neural network architecture significantly influences fraud detection performance. Overall, the results underscore the effectiveness of tailored enhancements such as weighted neighbors and common neighbors in improving the detection of fraudulent transactions within large-scale datasets.

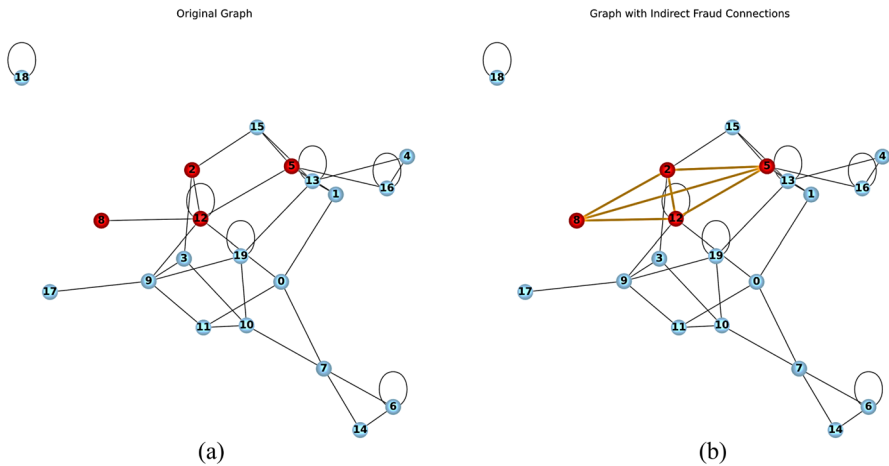
#### 4.6 Experiments on benchmark datasets

The evaluation results of the proposed methods and baseline methods on the two benchmark datasets, YelpChi and Amazon, are presented in Table 8. The proposed methods (proposed-method-WN and proposed-method-WN-CN) are compared with various GNN-based methods, such as GCN, GAT, GraphSAGE, DR-GCN, GPRGNN, FAGCN, GraphConsis, CARE-GNN, PC-GNN, Rio-GNN, AO-GNN, H2-FDetector, and CIES-GNN [7], to assess their performance in fraud detection.

In these experiments, the datasets were split into training, testing, and validation sets with proportions of 40%, 40%, and 20%, respectively. This ensures a balanced

**Table 9** Ablation study results on YelpChi and amazon datasets

| Metrics                          | YelpChi AUC | YelpChi F1-macro | YelpChi Gmean | Amazon AUC | Amazon F1-macro | Amazon Gmean |
|----------------------------------|-------------|------------------|---------------|------------|-----------------|--------------|
| Proposed-method-WN (Full)        | 0.8952      | 0.7325           | 0.8065        | 0.9705     | 0.8652          | 0.9220       |
| WN without attention             | 0.8801      | 0.7214           | 0.7913        | 0.9642     | 0.8520          | 0.9155       |
| WN without spatial-temporal Aggr | 0.8745      | 0.7152           | 0.7808        | 0.9598     | 0.8485          | 0.9087       |
| WN without GAE reconstruction    | 0.8650      | 0.7055           | 0.7723        | 0.9571     | 0.8400          | 0.9023       |



**Fig. 6** **a** Original graph structure and **b** Representation of the hidden indirect connections between the fraudulent nodes

and rigorous evaluation process, helping to prevent overfitting and ensure that the model generalizes well to unseen data.

According to the Table 8, the proposed-method-WN approach provides better output for the YelpChi dataset than the Amazon dataset compared to the CIES-GNN approach.

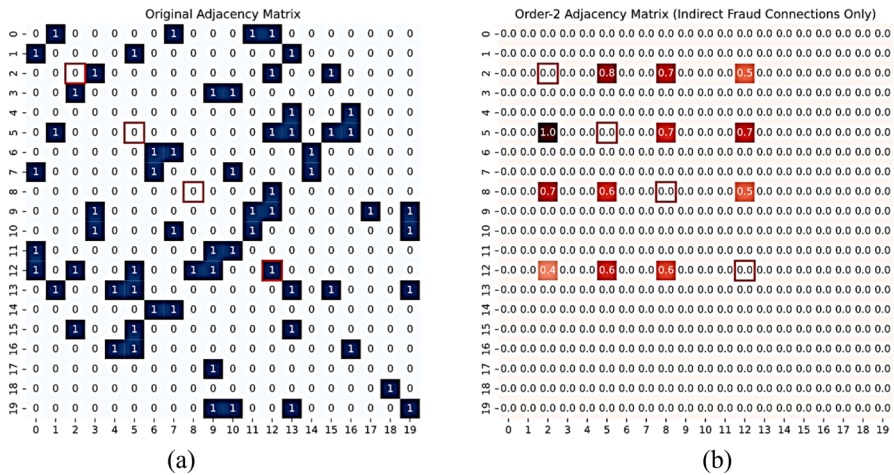
Considering that the YelpChi dataset has more fraudulent samples than the Amazon dataset and also has a reduced number of edges, the WN approach allows better detection of fraudster relations. In the adjacency matrix, instead of zero, the value of probabilistic relations between fraudulent nodes is replaced, thus the intensity of connections has been considered. Therefore, the WN method works better for the YelpChi dataset than the CIES-GNN approach for the Amazon dataset.

For the Amazon dataset, because the number of edges is higher than YelpChi and the number of fraudulent nodes has decreased, the CN method can better detect normal samples, and thus, it performs better than the CIES-GNN approach for the Amazon dataset.

The comparison results of evaluation criteria values for the proposed method and the basic methods on the Amazon and YelpChi benchmark datasets are shown in Figs. 3 and 4, respectively.

#### 4.7 Evaluation of hyperparameter sensitivity in long-path relations for fraud detection

In this section, we analyze the effects of different long-path relation values  $k$  during the graph reconstruction process. The results of this investigation are shown in Fig. 5, showcasing the performance metrics—AUC, F1-macro, and Gmean—for both the proposed-method-WN and proposed-method-WN-CN.



**Fig. 7** **a** Original adjacency matrix and **b** strength of the indirect fraud connections in the adjacency matrix

Our findings reveal that varying  $k$  significantly impacts the detection capabilities of our models. Specifically, for the Proposed-method-WN, the optimal performance was achieved for  $k = 5$ , yielding an AUC of 0.8952, an F1-macro of 0.7325, and a Gmean of 0.8065. In contrast, the proposed-method-WN-CN exhibited its best performance at  $k = 5$  as well, with an AUC of 0.9001, an F1-macro of 0.7336, and a Gmean of 0.8105.

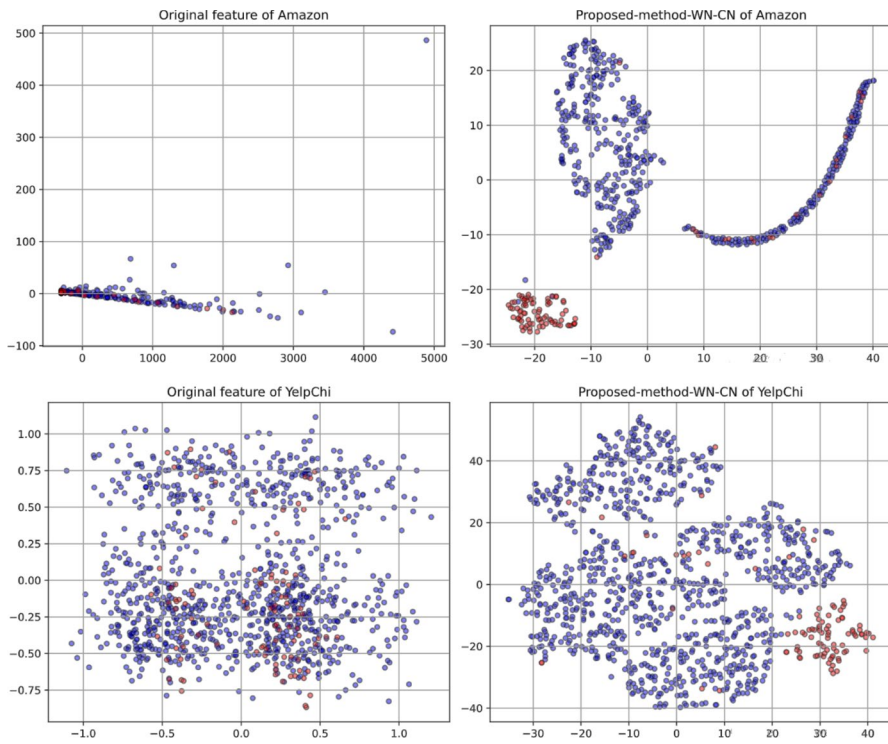
The results demonstrate that increasing the number of long-path relations enhances the model's ability to capture complex relationships within the data, thereby improving overall performance. However, as  $k$  increases beyond the optimal point, a decline in performance metrics is observed, indicating potential overfitting or diminishing returns from additional long-path relations. These insights underscore the importance of fine-tuning hyperparameters to achieve optimal fraud detection outcomes.

#### 4.8 Ablation study on module contributions in fraud detection

To evaluate the contribution of key components in the proposed method, we conducted an ablation study by systematically removing each of the main modules: attention mechanism, spatial-temporal aggregation, and GAE reconstruction. The results of the ablation study are presented in Table 9, where the full method (proposed-method-WN) is compared to its variants with individual components removed.

**Full Proposed-method-WN:** The method with all components included achieves the highest performance, with an AUC of 0.8952 on the YelpChi dataset and 0.9705 on the Amazon dataset. The F1-macro and Gmean metrics also reflect superior





**Fig. 8** Nodes embedding distribution

performance for both datasets (F1-macro: 0.7325, Gmean: 0.8065 for YelpChi; F1-macro: 0.8652, Gmean: 0.922 for Amazon).

**Without Attention Mechanism:** Removing the attention mechanism leads to a performance drop, with the AUC decreasing to 0.8801 for YelpChi and 0.9642 for Amazon. Similarly, the F1-macro and Gmean scores are reduced for both datasets. This demonstrates that the attention mechanism significantly contributes to improving the model's ability to focus on important features and relationships in fraud detection.

**Without Spatial–Temporal Aggregation:** The removal of the spatial–temporal aggregation module also leads to a decrease in performance, with AUC values dropping to 0.8702 for YelpChi and 0.9581 for Amazon. This highlights the importance of capturing both spatial and temporal patterns in the data for effective fraud detection.

**Without GAE Reconstruction:** Excluding the GAE reconstruction module results in a notable reduction in all metrics. The AUC decreases to 0.8720 for YelpChi and 0.9625 for Amazon, while both F1-macro and Gmean show a similar decline. This indicates that GAE reconstruction plays a critical role in enhancing the model's representational capabilities by reconstructing graph structures and preserving important node relationships.

Overall, the ablation study clearly shows that each module contributes significantly to the overall performance, with the attention mechanism, spatial-temporal aggregation, and GAE reconstruction being essential for achieving optimal results in fraud detection.

## 4.9 Visualization of graph reconstruction

In this subsection, we examine the importance of graph reconstruction in the fraud detection process. Figure 6 presents an example of several fraudulent and benign nodes from the real-world dataset. Nodes 2, 5, 8, and 12 are marked as fraudulent, while the remaining nodes are considered benign. In Fig. 6a, the original graph structure is shown, while Fig. 6b highlights the hidden indirect connections between the fraudulent nodes. Through the process of graph reconstruction, these indirect connections among fraudulent nodes are identified and incorporated into the graph.

Additionally, in Fig. 7b, the strength of the indirect connections between fraudulent nodes is illustrated. These connection strengths are determined based on the similarities identified among the fraudulent nodes and are integrated into the adjacency matrix, subsequently becoming part of the final reconstructed graph.

### 4.9.1 Visualization analysis of node distribution

This section presents a visual analysis of the distribution between fraudulent and benign nodes based on their original feature representations and the final layer node vector representations for the proposed method. The results, as shown in Fig. 8, indicate that using the proposed-method-WN-CN within the attention mechanism allows the final node embeddings of our method to effectively differentiate between the two types of nodes.

### 4.9.2 Statistical test

In this section, we performed statistical tests to compare and examine the performance of the proposed method. Based on fivefold cross-validation, the statistical Friedman test approach has been employed for this purpose. The Friedman test result showed statistically significant differences between the performances of the evaluated GNN-based models with ACC:  $F=27.90$ ,  $P=0.01$ ; F1-macro:  $F=23.70$ ,  $P=0.04$ ; and Gmean:  $F=26.05$ ,  $P=0.02$ . The proposed method can be selected as the best model since it provided the best performance for fraud detection.

## 5 Conclusion

In this paper, we introduced a novel method for detecting fraudulent transactions using GNNs, which leverages spatial-temporal information, preserves structural information, and considers potential relations to enhance detection accuracy. The proposed method incorporates two key innovations: attentional intra-relation and inter-relation aggregation techniques. These enhancements enable the method

to outperform traditional GNN architectures, such as GCN and GAT, as well as advanced methods like CIES-GNN, on both real-world and benchmark datasets.

The experimental results demonstrated that the proposed method achieves superior performance in terms of AUC, F1-macro, and Gmean metrics. Notably, it showed consistent improvements across diverse datasets, including the Yelp-Chi and Amazon datasets, by effectively addressing challenges posed by varying data densities and fraud sample distributions. For instance, the proposed-method-WN-CN achieved a 2.1% improvement in AUC over CIES-GNN on the Amazon dataset and a 1.2% improvement on the YelpChi dataset, demonstrating its adaptability to different fraud detection datasets.

Compared to previous methods, the proposed approach offers several key advantages. It enhances discrimination power by leveraging weighted neighbors (WN) and common neighbors (CN), effectively capturing subtle relationships that traditional models often overlook. Additionally, it demonstrates scalability and robustness, performing well on datasets with varying edge densities and fraud distributions, which highlights its ability to generalize to different environments. Furthermore, the integration of attention mechanisms enables the model to prioritize significant relationships, resulting in more accurate fraud detection.

The validity of this approach is supported by its consistent performance across datasets and its ability to address challenges associated with fraud detection, such as class imbalance and sparse connections. These strengths make the proposed method a promising solution for large-scale, real-world fraud detection applications.

In future work, we aim to explore the integration of advanced deep learning models, such as convolutional neural networks (CNNs), into the proposed framework to further enhance its performance. Additionally, expanding the method to handle dynamic graphs and incorporating domain-specific knowledge could lead to even more robust and accurate fraud detection systems.

**Author contributions** S. KH. and M. K. contributed to conceptualization of the study and methodology; S. KH. contributed to software and formal analysis; M. K. contributed to data curation and supervision; S. KH., B.T., and M.T. contributed to writing—original draft; B. T. and M. T. contributed to investigation, resources, review, and editing.

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**Data availability** No datasets were generated or analyzed during the current study.

## Declarations

**Conflict of interest** The authors declare no competing interests.

## References

1. Xie Y, Liu G, Yan C, Jiang C, Zhou M, Li M (2022) Learning transactional behavioral representations for credit card fraud detection. *IEEE transactions on neural networks and learning*

- systems. *IEEE Trans Neural Netw Learn Sys* 35:5735–5748. <https://doi.org/10.1109/TNNLS.2022.3208967>
2. The Nilson Report. [Online]. Available: <https://nilsonreport.com/mention/1313/1link>
  3. Cheng D, Wang X, Zhang Y, Zhang L (2020) Graph neural network for fraud detection via spatial-temporal attention. *IEEE Trans Knowl Data Eng* 34:3800–3813. <https://doi.org/10.1109/TKDE.2020.3025588>
  4. Wang D, Lin J, Cui P, Jia Q, Wang Z, Fang Y, Yu Q, Zhou J, Yang S, Qi Y (2020) A semi-supervised graph attentive network for financial fraud detection. In: 2019 IEEE International Conference on Data Mining (ICDM) 598–607. <https://doi.org/10.1109/ICDM.2019.00070>
  5. Liu C, Sun L, Ao X, Feng J, He Q, Yang H (2021) Intention-aware heterogeneous graph attention networks for fraud transactions detection. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining, 3280–3288. <https://doi.org/10.1145/3447548.3467142>
  6. Zhong Q, Liu Y, Ao X, Hu B, Feng J, Tang J, He Q (2020) Financial defaulter detection on online credit payment via multi-view attributed heterogeneous information network. In: Proceedings of the Web Conference 785–795. <https://doi.org/10.1145/3366423.3380159>
  7. Zhang J, Lu J, Tang X (2024) Two-stage GNN-based fraud detection with camouflage identification and enhanced semantics aggregation. *Neurocomputing* 570:127108. <https://doi.org/10.1016/j.neucom.2023.127108>
  8. Dou Y, Liu Z, Sun L, Deng Y, Peng H, Yu PS (2020) Enhancing graph neural network-based fraud detectors against camouflaged fraudsters. In: Proceedings of the 29th ACM International Conference on Information & Knowledge Management 315–324. <https://doi.org/10.1145/3340531.3411903>
  9. Liu Z, Dou Y, Yu PS, Deng Y, Peng H (2020) Alleviating the inconsistency problem of applying graph neural network to fraud detection. In: Proceedings of the 43rd international ACM SIGIR Conference on Research and Development in Information Retrieval 4:1569–1572. <https://doi.org/10.1145/3397271.3401253>
  10. Liu Y, Ao X, Qin Z, Chi J, Feng J, Yang H, He Q (2021) Pick and choose: a GNN-based imbalanced learning approach for fraud detection. In: Proceedings of the Web Conference 3168–3177. <https://doi.org/10.1145/3442381.3449989>
  11. Shi F, Cao Y, Shang Y, Zhou Y, Zhou C, Wu J (2022) H2-fdetector: A GNN-based fraud detector with homophilic and heterophilic connections. In: Proceedings of the ACM Web Conference 1486–1494. <https://doi.org/10.1145/3485447.3512195>
  12. Liu Y, Sun Z, Zhang W (2023) Improving fraud detection via hierarchical attention-based graph neural network. *J Inform Sec Appl* 72:103399. <https://doi.org/10.1016/j.jisa.2022.103399>
  13. Wang X, Ji H, Shi C, Wang B, Ye Y, Cui P, Yu PS (2019) Heterogeneous graph attention network. In: the World Wide Web Conference 2022–2032. <https://doi.org/10.48550/arxiv.1903.07293>
  14. Chen J, Chen Q, Jiang F, Guo X, Sha K, Wang Y (2024) SCN\_GNN: a GNN-based fraud detection algorithm combining strong node and graph topology information. *Expert Syst Appl* 237:121643. <https://doi.org/10.1016/j.eswa.2023.121643>
  15. Wu J, Hu R, Li D, Ren L, Huang Z, Zang Y (2024) Beyond the individual: an improved telecom fraud detection approach based on latent synergy graph learning. *Neural Netw* 169:2031. <https://doi.org/10.1016/j.neunet.2023.10.019>
  16. Lu M, Han Z, Rao SX, Zhang Z, Zhao Y, Shan Y, Raghunathan R, Zhang C, Jiang J (2022) Bright-graph neural networks in real-time fraud detection. In: Proceedings of the 31st ACM International Conference on Information & Knowledge Management 3342–3351. <https://doi.org/10.1145/3511808.3557136>
  17. Singh K, Tsai YC, Li CT, Cha M, Lin SD (2023) GraphFC: Customs Fraud Detection with Label Scarcity. In: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management. 4829–4835. <https://doi.org/10.1145/3583780.3614690>
  18. Van Belle R, Van Damme C, Tytgat H, De Weerd J (2022) Inductive graph representation learning for fraud detection. *Expert Syst Appl* 193:116463. <https://doi.org/10.1016/j.eswa.2021.116463>
  19. Zeng Y, Tang J (2021) RLC-GNN: An improved deep architecture for spatial-based graph neural network with application to fraud detection. *Appl Sci* 11:5656. <https://doi.org/10.3390/app11125656>

20. Rao SX, Zhang S, Han Z, Zhang Z, Min W, Chen Z, Shan Y, Zhao Y, Zhang C (2020) xFraud: explainable fraud transaction detection. arXiv preprint [arXiv:2011.12193](https://arxiv.org/abs/2011.12193). <https://doi.org/10.14778/3494124.3494128>.
21. Li Y, Cao J, Xu Y, Zhu L, Dong ZY (2024) Deep learning based on transformer architecture for power system short-term voltage stability assessment with class imbalance. *Renew Sustain Energy Rev* 189:113913. <https://doi.org/10.1016/j.rser.2023.113913>
22. Kim GB, Kim JY, Lee JA, Norsigian CJ, Palsson BO, Lee SY (2023) Functional annotation of enzyme-encoding genes using deep learning with transformer layers. *Nat Commun* 14:7370. <https://doi.org/10.1038/s41467-023-43216-z>
23. Fu S, Gao X, Zhai F, Li B, Xue B, Yu J, Meng Z, Zhang G (2024) A time series anomaly detection method based on series-parallel transformers with spatial and temporal association discrepancies. *Inf Sci* 657:119978. <https://doi.org/10.1016/j.ins.2023.119978>
24. An H, Ma R, Yan Y, Chen T, Zhao Y, Li P, Li J, Wang X, Fan D, Lv C (2024) Finsformer: a novel approach to detecting financial attacks using transformer and cluster-attention. *Appl Sci* 14:460. <https://doi.org/10.3390/app14010460>
25. Mao X, Liu M, Wang Y (2022) Using GNN to detect financial fraud based on the related party transactions network. *Procedia Computer Science* 214:351–358. <https://doi.org/10.1016/j.procs.2022.11.185>
26. Meng L, Ren Y, Zhang J (2023) Decoupling Graph Neural Network with Contrastive Learning for Fraud Detection. In: *International Conference on Database Systems for Advanced Applications* 397–414. [https://doi.org/10.1007/978-3-031-30678-5\\_30](https://doi.org/10.1007/978-3-031-30678-5_30).
27. Tian Y, Liu G (2023) Transaction fraud detection via spatial-temporal-aware graph transformer arXiv preprint arXiv: 2307. <https://doi.org/10.48550/arXiv.2307.05121>.
28. Hu S, Zhang Z, Luo B, Lu S, He B, Liu L (2023) BERT4ETH: a pre-trained transformer for ethereum fraud detection. In: *Proceedings of the ACM Web Conference* 2189–2197. <https://doi.org/10.1145/3543507.3583345>.
29. Tang Y, Liu Z (2024) A distributed knowledge distillation framework for financial fraud detection based on transformer. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2024.3387841>
30. Fan Y, Yeh CC, Chen H, Zheng Y, Wang L, Wang J, Dai X, Zhuang Z, Zhang W (2023) Spatial-temporal graph boosting networks: enhancing spatial-temporal graph neural networks via gradient boosting. In: *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management* 504–513. <https://doi.org/10.1145/3583780.3615066>.
31. Chen JP, Lu P, Yang F, Chen R, Lin K (2022) Medical insurance fraud detection using graph neural networks with spatio-temporal constraints. *J Netw Intell* 7:480–498
32. Hu B, Zhang Z, Shi C, Zhou J, Li X, Qi Y (2019) Cash-out user detection based on attributed heterogeneous information network with a hierarchical attention mechanism. *Proceed AAAI Conf Artif Intell* 33:946–953. <https://doi.org/10.1609/AAAI.V33I01.3301946>
33. Liu Z, Chen C, Li L, Zhou J, Li X, Song L, Qi Y (2019) Geniepath: Graph neural networks with adaptive receptive paths. *Proceed AAAI Conf Artif Intell* 33:4424–4431. <https://doi.org/10.1609/aaai.v33i01.33014424>
34. Motie S, Raahemi B (2023) Financial fraud detection using graph neural networks: A systematic review. *Expert Syst Appl* 204:122156. <https://doi.org/10.1016/j.eswa.2023.122156>

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